

Älyviitta 2.0

Technical Concept of an Open System for Warning of Wildlife Entering the Road

Document Status: Working engineering concept. Parameters marked as targets or assumptions require laboratory and field validation.

1. Mission and Principle

Älyviitta 2.0 is an open engineering initiative aimed at reducing vehicle collisions with large wild animals, primarily moose and reindeer.

The core idea is a distributed network of autonomous roadside beacons that detects hazardous animal movement, processes data locally, and generates a warning for the driver. The network is intended to continue performing its safety function even in the absence of mobile connectivity.

Key principle: the system should assess not merely the presence of a warm object, but the probability that an animal will enter a hazardous road zone within the next few seconds.

2. System Architecture

The system consists of repeating nodes installed along the road. Nodes may perform three logical roles: MASTER — sensor-equipped node; RELAY — intermediate node for radio transmission; SLAVE — visual warning node. Physically, a single hardware platform may combine multiple roles.

The MASTER receives data from the thermal array, motion sensor, ambient-light sensor and, when required, a ranging sensor. The local processor estimates direction, speed and detection confidence.

An 868 MHz radio network provides communication between neighboring nodes. NB-IoT/LTE-M is used primarily for remote telemetry, diagnostics and maintenance, rather than as a mandatory part of the local safety loop.

3. Wildlife Detection

The baseline sensor block consists of a Grid-EYE 8×8 thermal array, motion sensor and ambient-light sensor. The thermal array should be treated as a source of a spatiotemporal thermal map, rather than as a full thermal imaging camera.

To improve classification accuracy, an additional ToF/LiDAR module is proposed — a ranging sensor based on time-of-flight or laser measurement. Its purpose is to estimate distance, geometry and changes in the object's position.

Without ToF/LiDAR, the baseline system should be described as a thermal-event detection and motion-estimation system, rather than as a guaranteed species or mass classifier.

The algorithm must account for false triggers caused by vehicles, people, machinery, solar heating and other heat sources. A real winter-condition dataset must be collected.

4. Dynamic Hazard Zone

Instead of a fixed alarm circle, a dynamic risk zone is used. Its position and size change depending on the animal's direction of travel, estimated speed, distance from the road and site geometry.

When confidence is high, nodes near the presumed direction of the animal's road entry receive a red warning command. Lateral nodes may display a green wave that directs the driver's attention along the road section.

For uncertain cases, an intermediate yellow/amber state should be considered. This makes it possible to distinguish a confirmed hazard from an event requiring additional verification.

5. Modular Beacon Construction

The concept retains a vertical modular architecture: CIGS solar shell; transparent polycarbonate tube with bidirectional RGB LEDs; electronic module; LiFePO₄ battery; flexible connecting element; composite ground spike.

Module connections should use a bayonet-type mechanical latch with a spring retainer, witness mark, and electrical contacts via spring-loaded contacts or a concentric ring. Sealing is provided by an O-ring.

Cold resistance of the connection, fatigue strength of the flexible element, and sealing performance after repeated freeze/thaw cycles are currently subjects for validation.

6. Power System and Winter Autonomy

Power supply: flexible CIGS solar panel, LiFePO₄ battery, charge controller and a buffer supercapacitor for pulse loads.

The target operating mode is adaptive reduction of energy consumption below -20°C while maintaining full response speed of the emergency channel.

The stated 45–60 day autonomous operation in complete darkness should be treated only as a design target until the energy budget is completed and climate-chamber testing is performed.

Power consumption must be measured separately in deep sleep, standby, radio transmission, detection and alarm states.

The hypothesis of partial passive battery heating from heat generated by the microcontroller and electronics requires experimental validation.

7. Radio Network and Fault Tolerance

The network is designed as a distributed radio system with message transmission between neighboring nodes. This allows an alarm to propagate along a road section even if one beacon fails.

Overlapping detection zones improve resilience to a single sensor failure, but must not be described as absolute redundancy or as a guarantee of zero blind spots.

A target latency of less than 20 ms per radio hop should be treated as a design requirement until confirmed by measurement.

8. Self-Diagnostics and Cybersecurity

Each MASTER performs a periodic Heartbeat — a status message covering power, sensors, the radio link and internal errors. When necessary, a faulty node is marked as problematic.

Control commands should include authentication, replay protection against old messages, firmware integrity verification, and secure software updates.

The local warning function must have a safe operating mode in the event of loss of mobile connectivity or the remote server.

9. Validation and Test Stages

Stage 1: laboratory testing of electronics, radio communications, power, sealing and mechanics.

Stage 2: low-temperature climate testing, freeze/thaw cycles, battery testing and pulse-load testing.

Stage 3: controlled testing on Arctic snowmobile trails or closed sections with simulated animal movement.

Stage 4: collection of real-world data on moose and reindeer, followed by training and validation of detection algorithms.

Stage 5: pilot deployment on a public road after baseline safety, reliability and warning usefulness have been confirmed.

10. What Is Established and What Requires Validation

Already formulated: distributed network architecture, modular construction, local warning principle, separation of local safety from remote telemetry, and the self-diagnostic concept.

Requires validation: actual detection range and usefulness of the Grid-EYE 8×8 at 100 m, classification quality, false-trigger rate, winter energy balance, 45–60 day autonomy, cold-weather LiFePO4/BMS performance, radio-network latency, antenna and optical icing, bayonet-connection strength, flexible-joint strength, and height above snowbanks.

The following should not yet be stated as facts: guaranteed absence of blind spots, determination of animal mass using an 8×8 thermal array, confirmed 45–60 day autonomy, or a serial cost of approximately €20/€41.

11. Open Repository Structure

Version 1.1 should be retained as a historical engineering version. Version 2.0 should not be inserted into the repository as arbitrary lines of program code.

Recommended structure:

```
smart-auraviitta-open-hardware/  
README.md  
docs/ALYVIITTA_2.0.md  
docs/ARCHITECTURE.md  
docs/VALIDATION_PLAN.md  
docs/POWER_BUDGET.md  
docs/field-tests/  
hardware/  
firmware/  
simulations/  
CHANGELOG.md  
LICENSE
```

Code should be added as a real prototype emerges: first the minimum firmware and sensor drivers, followed by the radio protocol, alarm logic, diagnostics and tests.

12. Core Principle of Älyviitta 2.0

Älyviitta 2.0 is not simply a roadside post with illumination. It is an autonomous distributed safety system:

detection → local processing → risk assessment → radio alarm propagation → visual warning → self-diagnostics → remote monitoring.

The final goal is to create an open, verifiable and scalable solution that can be adapted for Finland, Sweden, Norway, Canada and other regions with a high probability of vehicle–wildlife collisions.

Note: This document is an engineering concept and research plan, not a certified road-safety system specification.